

Shubhangi S. R. Garnaik

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LinkedIn • GitHub • Google Scholar • DBLP

Research profile. Ph.D. student developing human-centered, deployable AI across multimodal interaction, augmented reality, speech systems, and trustworthy AI, with an emphasis on performance, latency, robustness, and real-world usability.

Education

2024–Present	Ph.D. in Computer Science Stony Brook University Advisor: Professor Jihoon Ryoo
2022–2023	M.S. in Computer Science Stony Brook University GPA: 3.92/4.00; supported through university and advisor scholarships
2015–2019	Bachelor of Engineering in Computer Engineering Savitribai Phule Pune University GPA: 3.33/4.00

Research Experience

2024–Present	Graduate Researcher, AI2S Lab Stony Brook University - Develop multimodal conversational and augmented-reality systems with grounded responses, low-latency pipelines, reliability safeguards, and user-centered evaluation. - Current work includes LocalMate, an accent-adaptive voice agent, and ArtSpeak, an interactive mobile AR experience for museum portraits.
Jul.–Dec. 2025	Research Intern, MEIC Lab Stony Brook University Implemented and containerized a reproducible 3D Gaussian Splatting pipeline; designed backend-API edge-case tests and documented failure modes.
2025	Research Assistant, School–Company Collaboration Taeyang Metal Industrial Co., Ltd. Developed computer-vision workflows for real-time forging monitoring, production-video analysis, anomaly detection, and equipment-failure prediction.

Professional Experience

Mar.–Oct. 2024	Data Scientist Thanks Carbon, Republic of Korea Developed deep-learning models from weather and rice-yield data to improve agricultural efficiency and support lower-carbon alternate wetting and drying practices.
Jun. 2019–Sep. 2021	Software Test Engineer / Associate Technical Sales Engineer Zensoft / Qualitia, India Automated client software testing and created structured validation workflows to reduce testing time and improve software quality.

Technical Skills

Programming: Python, Java, C#, HTML/CSS **AI/ML and Vision:** PyTorch, TensorFlow, OpenCV, deep learning, computer vision, video analytics, 3D Gaussian Splatting, snnTorch, Lava

Systems and Practice: Git, Linux, Docker, Jupyter, Google Colab, Unity, Android Studio, VS Code, Eclipse, backend API testing, test automation, reproducible workflows, technical documentation

Teaching

2025–2026	Teaching Assistant Stony Brook University CSE 113: Foundations of Computer Science (Professor Zhoulai Fu); CSE 310: Computer Networks (Professor Sehwan Yoo); CSE 373: Analysis of Algorithms (Professor James Finn).
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Publications

2026	Interspeech, accepted	When Machines Speak Like Local Peers: Improving Conversational Experiences with Accent-Adaptive Voice Agents Shubhangi S. R. Garnaik , JiHyun Jeong, and Jihoon Ryoo paper • project page
2025	IEEE ISMAR	ArtSpeak: An Interactive AR Application for Lifelike Speaking with Art Portraits Shubhangi S. R. Garnaik , Aruna Balasubramanian, Niranjana Balasubramanian, and Jihoon Ryoo IEEE Xplore • paper • project page
2025	ACM TOMM	CLOUD-CODEC: A New Way of Storing Traffic Camera Footage at Scale Hoyoung Kim, Azimbek Khudoyberdiev, Shubhangi S. R. Garnaik , Arani Bhattacharya, and Jihoon Ryoo DOI: 10.1145/3744649
2023	IEEE Access	A Study of the Man and Unmanned Teaming System for Improving Efficiency in a Fulfillment Center Azimbek Khudoyberdiev, Shubhangi S. R. Garnaik , Ha Young Kim, and Jihoon Ryoo DOI: 10.1109/ACCESS.2023.3341396
2023	IEEE ICTC	MINSU: Precision Quantity Counter with DNN-based Volume Estimation Shubhangi S. R. Garnaik , DongYeop Lee, Youngho Kim, and Jihoon Ryoo DOI: 10.1109/ICTC58733.2023.10392667
2022	IEEE ICTC	SQR: Secure QR Transaction with Randomized Rotation Shubhangi S. R. Garnaik , Youngho Kim, and Jihoon Ryoo DOI: 10.1109/ICTC55196.2022.9952603

Posters and Workshop Papers

2023	Enhancing Liver Disease Prediction Using Machine Learning Models Comparative study of oversampled and synthetic-data approaches; presented at KICS.
2023	Exploring Sentiment Analysis of the Silicon Valley Bank Collapse Deep-learning approaches applied to Google News headlines; presented at KICS.

Honors and Awards

Winner, Decompiler HACKATHON THE X

May 2026

Idea Finalist, Flipkart Hackathon

2019

Conference Scholar, Distributed Systems Conference

2019

Service and Leadership

Orientation Leader supporting incoming students; organizer of an inter-college coding competition in Pune.

References

Professor Jihoon Ryoo

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Professor Aruna Balasubramanian

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